Analysis: overfitting, datasets and what the metrics mean

M1 CBD-Net and M2 Improved DeepLabV3+

1. Are the datasets a problem?

Table 1. SOS Sentinel-1: tiles vs independent sea areas.

Split	Train tiles	Test tiles	Independent sea areas (train)	Independent sea areas (test)	Areas in both train and test	Oil share train	Oil share test
Official (leaky)	3354	839	1089	444	444 (all test areas)	29.3 %	34.7 %
Group-disjoint (leak-free)	3358	835	877	212	0	30.6 %	29.6 %

- **SOS is much smaller than its tile count.** 4,193 Sentinel-1 tiles come from only 1,089 independent sea areas (an area appears up to 5 times, as augmented copies). The effective training set is therefore about a third of its nominal size, which raises the risk of memorising areas.
- **In the official split, every test area also appears in training**, so official test scores partly measure recognition of seen areas (the leak). The group split removes this.
- **The leak-free test set has less oil** (29.6 % vs 34.7 % of pixels), which raises background IoU, mIoU and accuracy by itself. Compare models on oil IoU and oil F1.

Table 2. Peru and MDOD (test only, never used for training).

Dataset	Size	Balance check	Effect on our numbers
Peru	2,013 patches, 40 scenes (median 41, min 4, max 161 patches per scene)	largest scene = 8.0 % of all patches	pooled metrics are weighted towards large scenes; we therefore bootstrap by scene, so the confidence intervals reflect 40 scenes, not 2,013 patches
MDOD	oil spill: 10,763 boxes; biogenic slick: 3,054 boxes; low wind: 4,695 boxes; rain cell: 4,658 boxes; sea ice: 1,770 boxes	we sample 400 boxes per class (seed 0), so every class has equal weight	oil spill median short side 20 px; biogenic slick median short side 330 px; low wind median short side 138 px; rain cell median short side 92 px; sea ice median short side 220 px

- **The large datasets are not a problem for training: they are never trained on.** They only test. Their size gives tighter confidence intervals. What matters is balance, which is handled by sampling equal numbers of boxes per class (MDOD) and by resampling whole scenes (Peru).
- **MDOD differs in scale:** its scenes are about 10× coarser than SOS tiles, so oil boxes contain thin streaks. This lowers the oil alert rate for both models and is a property of the dataset, not of overfitting.
- **MDOD also differs in brightness** (pixel brightness on a random sample (SOS 150 tiles, Peru 150 patches, MDOD 25 scenes, land excluded): mean / median SOS 96 / 89, Peru 88 / 85, MDOD 135 / 142). Peru matches SOS closely, so the Peru failure is not a brightness effect; MDOD mixes three effects (brightness, resolution, box size) and is therefore reported as a supplementary look-alike stress test, not as clean evidence.

2. Do the models overfit?

Table 3. Train vs test (single pass, no augmentation) and the external test. Train = the model's own training tiles.

Model	Train IoU oil	Test IoU oil	Gap	Train F1	Test F1	Gap	Train loss	Test loss	Loss ratio	Peru F1 (new sea)
M1, leaky-trained	78.19	78.09	+0.10	87.76	87.70	+0.07	0.3493	0.3785	1.08	24.49
M1, leak-free-trained	77.33	76.00	+1.33	87.22	86.37	+0.85	0.3807	0.3835	1.01	29.47
M2, leaky-trained	80.17	78.36	+1.81	89.00	87.87	+1.13	0.3160	0.3842	1.22	22.91
M2, leak-free-trained	80.60	76.91	+3.69	89.26	86.95	+2.31	0.3135	0.3621	1.16	22.34

Reading this table: a small train–test gap means the model generalises to new tiles from the same kind of data. A large drop from the SOS test to Peru means it does not generalise to a new sea. These are two different things: the first is classic overfitting, the second is domain shift.

- **M1 shows almost no overfitting.** Oil IoU gap +0.1 pp on the leaky split and +1.3 pp on the leak-free split. Its schedule stops when the training loss stops improving.
- **The near-zero gap of M1 on the official split is itself a symptom of the leak:** the official test tiles are augmented copies of training areas, so the test set behaves like the training set. On the leak-free split a gap appears.
- **M2 overfits mildly.** Oil IoU gap +1.8 pp (leaky) and +3.7 pp (leak-free); the test loss is 16–22 % above the training loss. M2 trains for the full 80 epochs with no early stopping, and the last epoch is evaluated (its paper uses no validation set).
- **Removing the leak makes the gap larger for both models** (oil IoU gap M1: +0.1 → +1.3 pp, M2: +1.8 → +3.7 pp), which confirms that the official split hides part of the generalisation gap.
- **Overall, overfitting is small (under 4 pp) compared with the domain shift:** oil F1 falls from about 87 % on SOS to 22–29 % on Peru. The models are not memorising SOS tiles; they learned a rule ("dark area = oil") that does not hold on other seas. The ranking of the effects is: domain shift (≈ 60 pp) ≫ overfitting (≤ 4 pp) > leak (≈ 1–2 pp).
- Caveat: training-tile scores are computed without augmentation, and the SOS training set itself contains augmented copies.

3. Are the differences real? (confidence intervals)

Table 4. Paired bootstrap differences. "Clear" = the 95 % interval excludes zero. One training seed per model, so seed-to-seed variation is not included and the true uncertainty is larger.

Comparison	Test set	Metric	Difference [95 % CI]	Clear?
M2 − M1 (leaky-trained)	SOS (same tiles)	F1	+0.14 [-0.54, +0.71] pp	no
M2 − M1 (leaky-trained)	SOS (same tiles)	IoU oil	+0.22 [-0.87, +1.13] pp	no
M2 − M1 (leaky-trained)	SOS (same tiles)	mIoU	+0.41 [-0.46, +1.14] pp	no
M2 − M1 (leak-free-trained)	SOS (same tiles)	F1	+0.41 [+0.07, +0.76] pp	yes
M2 − M1 (leak-free-trained)	SOS (same tiles)	IoU oil	+0.65 [+0.11, +1.20] pp	yes
M2 − M1 (leak-free-trained)	SOS (same tiles)	mIoU	+0.56 [+0.15, +0.97] pp	yes
M1: leak-free − leaky	Peru (same patches, by scene)	F1	+4.98 [+2.53, +6.60] pp	yes
M2: leak-free − leaky	Peru (same patches, by scene)	F1	-0.57 [-1.29, +0.07] pp	no
M2 − M1 (leaky-trained)	Peru (same patches, by scene)	F1	-1.58 [-2.38, -0.67] pp	yes

Comparison	Test set	Metric	Difference [95 % CI]	Clear?
M2 − M1 (leak-free-trained)	Peru (same patches, by scene)	F1	-7.13 [-8.95, -3.52] pp	yes

- **SOS, M2 vs M1:** on the leaky split the difference is not clear (F1 +0.14 pp [-0.54, +0.71]); on the leak-free split M2 is slightly but clearly better (F1 +0.41 pp [+0.07, +0.76]).
- **Peru, leaky vs leak-free training:** clear for M1 (F1 +4.98 pp [+2.53, +6.60]), not clear for M2 (-0.57 pp [-1.29, +0.07]). So removing the leak does not consistently improve transfer to a new sea.
- **Peru, M2 vs M1:** M1 is clearly better on the new sea (leak-free models: -7.13 pp [-8.95, -3.52]), the opposite of the SOS ranking. In-domain rankings do not predict out-of-domain rankings.
- **Peru absolute values are very uncertain:** e.g. M1 leaky-trained F1 24.5 % [10.2, 36.9]. Performance depends strongly on the scene, so a few scenes decide the score. Paired differences are much tighter than absolute values.
- **MDOD:** every AUC interval lies far below 0.5 (highest upper bound 0.37), so "look-alikes score higher than oil" is not chance.
- All intervals cover only test-set sampling. Each model is one training seed; seed-to-seed variation is extra.

4. What the metrics tell us

- **mIoU can mislead.** It averages oil IoU with background IoU, and background is about 70 % of the pixels. When the test set has less oil, background IoU rises and mIoU rises even if oil detection gets worse (leak-free split: oil IoU down, mIoU up). Oil IoU and oil F1 are the honest headline numbers; mIoU is reported because the literature uses it.
- **Accuracy is not informative here.** A model that predicts "no oil" everywhere scores about 70 % on SOS and about 94 % on Peru.
- **Precision vs recall tells the failure mode.** On Peru recall stays about 95 % but precision falls to 13–17 %: the models miss little oil but raise about 6 false alarms per true one. Operationally that is the costly error.
- **F1 vs IoU:** they always rank models the same way (IoU = F1 / (2 − F1)); IoU is the stricter number.
- **ROC-AUC vs ECE:** ROC-AUC stays high on Peru (≈ 0.91–0.93), meaning the probabilities still rank pixels sensibly, while ECE rises from ≈ 0.03 to 0.19–0.30, meaning the probabilities are no longer trustworthy as probabilities. That combination (good ranking, bad calibration) is exactly what calibration and threshold methods can fix.
- **MSE and R^2 (regression view of the probabilities):** on SOS R^2 is clearly positive, but on Peru it is strongly negative: the over-confident probabilities are worse than predicting the average oil share for every pixel. This is the same message as ECE.
- **Boundary F1 is low (38–48 %)** because SOS labels are coarse polygons; it measures label agreement on edges as much as model quality.
- **The official-script F1 (93–95 %) is a background-class score** and should never be quoted as oil F1.
- **MDOD scores are confounded by box size:** oil boxes are small (median short side 20 px) while look-alike boxes are large (92–330 px). Mean-score and coverage comparisons between classes are therefore partly comparisons of box size; the alert rate and AUC should be read as "how often the model raises an alarm on each phenomenon", not as pixel accuracy.

Metric definitions (same order as every table)

Table 5. Metrics. TP / FP / FN / TN = oil pixels predicted correctly / false oil / missed oil / correct background. Pixels are pooled over all test tiles. Higher is better except ECE and MSE.

Metric	Short	What it measures	Formula	Good value
Accuracy	Acc	share of all pixels classified correctly (inflated by the large sea background)	(TP+TN)/all	high, but not informative alone
Precision (oil)	P	of the pixels called oil, how many are oil (false-alarm control)	TP/(TP+FP)	high
Recall (oil)	R	of the real oil pixels, how many were found (missed-oil control)	TP/(TP+FN)	high
F1 (oil)	F1	balance of precision and recall; equals the Dice coefficient	2PR/(P+R)	high
IoU (oil)	IoU oil	overlap of predicted and true oil areas; stricter than F1	TP/(TP+FP+FN)	high
IoU (background)	IoU bg	same for the background class	TN/(TN+FP+FN)	high
mIoU	mIoU	mean of oil IoU and background IoU; the main SOS benchmark number. Rises when the test set has more background	(IoU oil + IoU bg)/2	high
Boundary F1	BF1	F1 of the slick outlines, matched within 2 pixels	—	high
F1, official script	F1 off.	what the official SOS script reports: F1 of the BACKGROUND class (a bug in that script)	2·Pbg·Rbg/(Pbg+Rbg)	for comparison only
ROC-AUC	ROC-AUC	how well probabilities rank oil pixels above non-oil, at any threshold	area under ROC	1.0 (0.5 = chance)
PR-AUC	PR-AUC	same, focused on the oil class; sensitive to class imbalance	area under precision-recall	high
ECE	ECE	calibration: gap between predicted probability and how often it is right (15 bins)	$\Sigma\ (n_b/N)\lvert acc_b - conf_b\rvert$	0
MSE (= Brier score)	MSE	mean squared error between predicted oil probability p and true label y (0/1) of every pixel	mean $(p - y)^2$	0
R^2	R^2	share of the label variance explained by the probabilities; 0 = no better than predicting the average oil share, negative = worse	1 − MSE / (f(1 − f)), f = oil share	1
TTA		test-time augmentation: average of 8 rotated / flipped predictions		
Alert rate (MDOD)		share of boxes in which at least x % of pixels have p(oil) > 0.5		high for oil, low for look-alikes
AUC oil vs class (MDOD)		chance that a random oil box scores higher than a random box of that class	Mann-Whitney	1.0 (0.5 = chance)